MATH 350-2 Advanced Calculus

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Outline

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Stieltjes sums

Let f and α be real-valued functions on $[a, b]$, and let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$ and define

$$m_k(f) = \inf\{f(x) : x_{k-1} \leq x \leq x_k\}, \ \ M_k(f) = \sup\{f(x) : x_{k-1} \leq x \leq x_k\}.$$

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and the **lower Stieltjes sum** of f with respect to α on $[a, b]$ is:

$$\underline{S}(P, f, \alpha) = \sum_{k=1}^{n} m_k(f)\Delta\alpha_k(P)$$

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These sums refine predictably!

Lemma

Let f and α be real-valued functions on $[a, b]$, with α monotone increasing.

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Lemma

Let f and α be real-valued functions on $[a, b]$, with α monotone increasing. If P is a partition of $[a, b]$ and P' is a refinement of P, then

$$\underline{S}(P, f, \alpha) \leq \underline{S}(P', f, \alpha) \leq \overline{S}(P', f, \alpha) \leq \overline{S}(P, f, \alpha)$$

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Proof.

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Let $P = \{x_0, \dots, x_n\}$ and $P' = \{x'_0, \dots, x'_N\}$.

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Let $P = \{x_0, \dots, x_n\}$ and $P' = \{x_0', \dots, x_N'\}$.
For each k, let $\iota(k)$ with $x_{\iota(k)-1} \leq x_k' \leq x_{\iota(k)}$.

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For each k, let $\iota(k)$ with $x_{\iota(k)-1} \leq x'_k \leq x_{\iota(k)}$.

$$m_{\iota(k)}(f) = \inf\{f(x) : x_{\iota(k)-1} \leq x \leq x_{\iota(k)}\} \leq \inf\{f(x) : x'_{k-1} \leq x \leq x'_k\} = m'_k(f)$$

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We will prove $\underline{S}(P, f, \alpha) \leq \underline{S}(P', f, \alpha)$.
Let $P = \{x_0, \dots, x_n\}$ and $P' = \{x_0', \dots, x_N'\}$.
For each k, let $\iota(k)$ with $x_{\iota(k)-1} \leq x_k' \leq x_{\iota(k)}$.

$$m_{\iota(k)}(f) = \inf\{f(x) : x_{\iota(k)-1} \leq x \leq x_{\iota(k)}\} \leq \inf\{f(x) : x_{k-1}' \leq x \leq x_k'\} = m_k'(f)$$

Therefore

$$\begin{aligned} \underline{S}(P', f, \alpha) &= \sum_{k=1}^{N} m_k'(f)\Delta\alpha_k(P') \\ &\geq \sum_{k=1}^{N} m_{\iota(k)}(f)\Delta\alpha_k(P') \\ &= \sum_{k=1}^{n} m_k(f)\Delta\alpha_k(P) = \underline{S}(P, f, \alpha) \end{aligned}$$

$\square$

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The **lower Stieltjes integral** of f with respect to α on $[a, b]$ is:

$$\underline{\int} f d\alpha = \sup\{\underline{S}(P, f, \alpha) : P \text{ partition of } [a, b]\}$$

Riemann's criterion

Definition

Let f and α be real-valued functions on $[a, b]$.
Then f **satisfies Riemann's criterion** with respect to α on $[a, b]$ if for all $\epsilon > 0$ there exists a partition P_ϵ such that for all refinements P of P_ϵ

$$0 \leq \overline{S}(P, f, \alpha) - \underline{S}(P, f, \alpha) < \epsilon.$$

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Theorem

Let f and α be real-valued functions on $[a, b]$, with α monotone increasing. Then the following are equivalent:
(a) f is Riemann integrable with respect to α on $[a, b]$
(b) f satisfies Riemann's criterion with respect to α on $[a, b]$
(c) $\underline{\int} f d\alpha = \overline{\int} f d\alpha$

Challenge!

Suppose that $f(x) = x$, $\alpha(x) = x$, and $[a, b] = [0, 1]$.

Problem

Calculate the value of

$$\overline{S}(P, f, \alpha) \text{ for } P = \{0/n, 1/n, \dots, n/n\}.$$

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Prove that $\int_0^1 f d\alpha = 1/2$.

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Suppose that $f(x) = x^2$, $\alpha(x) = x$, and $[a, b] = [0, 1]$.

Problem

Prove that $\int_0^1 f d\alpha = 1/3$. Hint:

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Continuous implies integrable

Theorem

Suppose that f is a continuous real-valued function on $[a, b]$. Then f is Riemann integrable on $[a, b]$.

Mean-Value Theorem for Integrals

Theorem

Suppose that f is integrable with respect to α on $[a, b]$. Let

$$m = \inf\{f(x) : x \in [a, b]\}, \quad \text{and} \quad M = \sup\{f(x) : x \in [a, b]\}.$$

Then there exists $m \leq c \leq M$ with

$$\int_a^b f d\alpha = c(\alpha(b) - \alpha(a)).$$

In the special case f is continuous, then $c = f(x_0)$ for some $x_0 \in [a, b]$.

Fundamental Theorem of Calculus I

Assume that f is integrable with respect to α on $[a, b]$ and set

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$$F(x) = \int_a^x f d\alpha, \quad x \in [a, b].$$

Theorem (FTC I)

Suppose that α is monotone increasing on $[a, b]$.
Then for any $x \in (a, b)$ where f is continuous and α is differentiable

$$F'(x) = f(x)\alpha'(x).$$

Fundamental Theorem of Calculus II

Theorem (FTC II)

Suppose that f is Riemann integrable on $[a, b]$.
Also suppose g is continuous on $[a, b]$ with $g'(x) = f(x)$ for all $x \in (a, b)$.
Then

$$\int_a^b f(x)dx = g(b) - g(a).$$